ITA03-MOBILE COMPUTING
ASSIGNMENT-1

ASSIGNMENT 1 – CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

Nature of Assignment: Individual
Duration: 1 Week
Total Marks: 50
Submission Date: 12/11/26

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon). ● Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.
- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km^2 stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

Evidence Expected
Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

TASK 2 – Identify the Governing Design Parameter 2 Marks Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).

- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km². ● Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each

blocking behaviour.

● analyse how shadowing and a changing path-loss exponent shift each case. ● Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

TASK 5 – Scaling and Sensitivity Analysis 3 Marks Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical?

Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure. ● Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

TASK 7 – Solve the Recurrence and Interpret the Result 4 Marks Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility?

Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4 (D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

TASK 8 – Implement and Test the Planning Strategies 4 Marks Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator. ● Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

TASK 9 – Record and Present Experimental Performance 3 Marks Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies. **Evidence**

Expected

Systematically collected data presented in comparable tables and graphs.

TASK 10 – Compare Theory with Simulation 4 Marks Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy. ● Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results. **Evidence**

Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 11 – Recommend and Defend the Final Design 4 Marks Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.

- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km^2 stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.

You may:

RETAIN → MODIFY → or REDESIGN

your original solution.

A. Impact Analysis – 2 Marks

Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density. ● Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2 Marks

Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your

own model.

● Assess whether the design can be reused unchanged and what fails first. ● Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge. **C.**

Adaptation or Redesign – 2 Marks

Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

● Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing). ● Describe the revised strategy and what changes in reuse pattern, power and handoffs.

● Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect. **D.**

Evidence-Based Defence – 2 Marks

Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

● Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.

● Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

ASSIGNMENT-1 – ANSWERS

Planning assumptions used throughout the answers

The township area is 12 km² and the available spectrum is fixed at 336 duplex traffic channels. Busy-hour offered traffic per subscriber is 0.025 Erlang and the target blocking probability is 2%. I select a minimum SIR target of 18 dB because it gives a conservative interference margin. For open campus areas I use path-loss exponent n = 2.5–3; for dense built-up areas I use n = 4. Unless otherwise stated, calculations use N = 7, six first-tier co-channel interferers, and the standard hexagonal-cell approximation. Numerical site and cost values are planning estimates, not vendor quotations.

TASK 1 – Design Alternative Frequency-Planning Strategies (5 Marks)

Strategy 1: Fixed cluster frequency reuse

The 336 traffic channels are divided into N channel groups and the groups are repeated across a hexagonal grid. With N = 7, each cell receives S/N = 336/7 = 48 channels. A cluster contains N cells and the same channel group is reused only after the reuse distance D. This is spatial reuse and is appropriate when traffic is relatively stable and interference must be predictable.
Hexagonal cluster sizes must satisfy $N = i^2 + ij + j^2$, where i and j are non-negative integers. The procedure below searches valid cluster sizes and chooses the smallest one meeting the SIR target.
Pseudocode:
Input: total_channels S=336, SIR_target=18 dB, path_loss n, cell radius R
Generate candidate N = i*i + i*j + j*j for integer i,j
For each N: channels_per_cell = floor(S/N)
Compute D/R = sqrt(3*N) and SIR = (D/R)^n / 6
Reject candidates whose SIR is below the target
Choose the smallest remaining N with adequate capacity; assign each channel group to N cells
Repeat the channel groups over the hexagonal grid
Preprocessing: prepare a subscriber/traffic heat map, measure propagation and shadowing, select a hexagonal grid, and verify candidate sites. The operator should pay for radio surveys, traffic modelling and site planning because these are part of network deployment.

Strategy 2: Sectorised cells

Each base station uses directional antennas instead of one omnidirectional antenna. A 120° layout has three sectors and a 60° layout has six sectors. Directional antennas reduce the number of dominant co-channel interferers seen by a sector. The design principle is spatial filtering. The channel pool is still fixed; channel groups are assigned to sectors so adjacent sectors and nearby co-channel sectors do not use interfering groups.
Pseudocode:
Input: hex grid, S=336, candidate N, sector count = 3 or 6
Divide each cell into sectors and assign channel groups by sector orientation
For sampled mobile positions, calculate desired signal and directional interference
Estimate the 5th-percentile SIR and outage probability

Increase N or change sector orientation if the SIR target is not met
Keep the sector plan with acceptable SIR and required capacity
Preprocessing: in addition to the traffic map and propagation survey, obtain antenna patterns, azimuths and tilt information. Sectorisation adds antenna and installation costs but can improve SIR without buying extra spectrum.

Strategy 3: Dynamic channel assignment / channel borrowing with hierarchical cell splitting

Channels are selected from a common eligible pool according to current interference and traffic rather than being permanently fixed to every cell. A lightly loaded cell may lend an unused channel to an overloaded neighbour if the reuse and interference constraints remain satisfied. For long-term growth, overloaded macro-cells are recursively split into smaller micro-cells. The principles are trunking efficiency, dynamic allocation and hierarchical reuse.
Pseudocode:
For each call request, list channels that are free and satisfy the interference constraint
Allocate a locally preferred channel; if none exists, request a safe channel from a neighbour
If no safe channel is available, block the call and record the event
At each planning interval, identify cells exceeding the traffic threshold
Split only overloaded cells, reduce radius, add sites and recompute channel assignment
Stop splitting when blocking is at most 2% or a cost/site limit is reached
Preprocessing: time-stamped busy-hour traffic, interference measurements, neighbour-cell data, candidate micro-site survey and a handoff map are needed. The operator pays for planning and the extra control software; borrowing can improve use of existing channels but requires coordination.

Access-channel interpretation

FDMA separates users by frequency; TDMA separates users by time slots; CDMA separates users by spreading codes; OFDMA assigns groups of orthogonal subcarriers and is common in modern cellular systems. The 336-channel figure is treated as the available duplex traffic-channel equivalent after control-channel overhead. Broadcast, synchronisation, paging and access/control resources must be reserved separately in a real system; they must not be counted again as traffic channels.

Technical Explanation and Application

Further explanation of fixed cluster reuse: The main purpose of a cluster is to organise the frequency pool so that neighbouring cells do not use the same group too closely. A cluster is repeated over the service region, and the same channel group is reused only after the required separation. This makes the channel plan predictable and easier to audit. The disadvantage is that a cell cannot automatically use every idle channel in the network. A cell may experience blocking even when another distant cell has unused channels. The plan is therefore strongest when traffic is reasonably stable and the cells have similar demand.

The choice of N must be considered together with the SIR requirement. A smaller cluster gives more channels to each cell, but it also places co-channel cells closer together. A larger cluster improves separation but reduces the number of channels allocated to each cell. For the given pool, N = 3 provides 112 channels per cell, N = 4 provides 84, N = 7 provides 48 and N = 12 provides 28. These values show why selecting the smallest cluster without checking interference can produce an unacceptable radio plan.

Sectorisation uses the antenna radiation pattern to control the direction of transmitted and received energy. A three-sector site divides the full horizontal coverage into three nominal 120-degree sectors. A six-sector site uses six nominal 60-degree sectors. The sector antenna does not remove interference completely; side lobes and reflections may still allow signals from other cells to reach a receiver. The practical benefit depends on antenna quality, azimuth, electrical tilt, mechanical tilt, site height and the location of neighbouring cells.

Dynamic channel assignment is different from fixed allocation because the network makes a channel decision when a call or connection request occurs. The controller checks whether a channel is free and whether using it will violate the interference rule. Channel borrowing is a related method in which an overloaded cell requests a channel that is currently unused in a neighbouring cell. The donor cell must retain enough resources for its own expected traffic. This method is useful when traffic varies by location or time, but it requires coordination and up-to-date channel occupancy information.

Hierarchical macro–micro planning uses more than one cell size. Macro-cells provide broad coverage over roads and less densely occupied parts of the township. Micro-cells are placed in locations where demand is concentrated, such as hostels, lecture buildings, libraries and the stadium. The smaller cells increase the number of locations at which the same spectrum can be reused. This approach is not simply a different cluster size: it changes the physical network layout and allows capacity to be added only where it is required.

The strategies can be combined because they address different limitations. Fixed reuse provides the basic frequency plan; sectorisation improves spatial separation; dynamic assignment improves the use of idle resources; and micro-cells increase local capacity. The preprocessing stage should therefore produce a traffic map, a propagation survey, a hexagonal planning grid, a list of candidate sites and an interference-neighbour table. These activities are operator deployment costs and should be included before equipment installation is approved.

Further design considerations

Channel allocation must also consider the difference between control and traffic resources. Control channels support access, paging, synchronisation and connection management, while traffic channels carry the user connection after assignment. The scenario states that 336 duplex traffic-channel equivalents remain after control-channel overhead. The calculations therefore use 336 directly as the traffic pool. Removing a second control-channel allowance would count the same overhead twice.

An important operational issue in fixed reuse is the treatment of cell boundaries. A mobile near a boundary may receive comparable power from two base stations. The serving cell should be selected using a suitable radio-quality and handoff rule, and the channel plan should be checked for conflicts in neighbouring cells. The ideal hexagonal model does not automatically determine

the serving cell in every real location because buildings and terrain can change the received signal.

For sectorisation, the channel groups should be coordinated between adjacent sectors and neighbouring sites. Assigning the same group to sectors that face one another may create strong interference even if the nominal site-to-site distance appears acceptable. Antenna orientation and tilt are therefore part of frequency planning, not merely installation details. The plan should be reviewed after commissioning because measured coverage may differ from the predicted pattern.

Dynamic assignment needs a decision rule that can be executed quickly when a request arrives. The controller can rank eligible channels by measured interference, reuse distance or a channel-quality score. It should also avoid repeatedly moving active calls between channels without a clear benefit. A stable allocation policy reduces unnecessary signalling and makes the network easier to operate during periods of changing demand.

The township plan should use the same terminology and comparison measures for all strategies. For every alternative, state the available channel pool, the reuse rule, the interference check, the expected capacity benefit and the additional cost. This makes it clear that the strategies are genuinely different methods of sharing a fixed resource, rather than simply different numerical choices of cluster size.

Practical Design Considerations

A useful way to compare the alternatives is to consider what happens when one cell becomes overloaded. Fixed reuse keeps its assigned group and may block calls even if another cell has spare channels. Dynamic borrowing can use some of that spare capacity if the channel is safe. Sectorisation may improve the radio conditions and allow more effective reuse. A micro-cell can provide additional local capacity when the overload is persistent. This comparison shows the practical difference between the methods.

The operator should also consider how easy each method is to maintain. Fixed reuse is relatively straightforward to document, while dynamic assignment needs a controller and reliable information about neighbouring cells. Sectorisation needs antenna configuration and optimisation. Micro-cells require site access, backhaul and power. These operational requirements should be included when judging whether a strategy is suitable for the township.

Before finalising a strategy, the planning team should draw the proposed cell layout and mark the channel groups, sectors and candidate micro-sites. The drawing should identify which cells are co-channel neighbours and where traffic hotspots are expected. A clear layout makes it easier to find conflicts and explain how the fixed channel pool is reused.

All alternatives must respect the same spectrum constraint. A strategy cannot claim additional licensed channels unless the scenario is changed. The capacity improvement must come from better spatial reuse, more efficient assignment, directional antennas or a denser layout. This is the central constraint that guides the entire assignment.

Implementation and planning notes

The frequency plan should be prepared as a map as well as a table. The map identifies the physical location of each cell and its neighbours, while the table records the channel group

assigned to each cell or sector. This helps the planning team identify adjacent co-channel assignments and makes the plan easier to verify during installation and later maintenance.

An important limitation of fixed reuse is that a fixed channel group cannot automatically follow a changing traffic hotspot. If a lecture hall becomes crowded during the day and quiet at night, its assigned resources may be insufficient at one time and underused at another. Dynamic assignment can respond to this variation, but it requires accurate and timely information about channel availability.

Sectorisation should be selected using the actual shape of the service area. A nominal 120-degree sector may not align perfectly with the road or building layout. Antenna azimuth can be adjusted so that the main beam covers the intended user area and reduces energy directed toward dominant co-channel interferers. The final orientation must be verified by measurements after installation.

Hierarchical planning is useful because macro and micro cells serve different purposes. Macro-cells maintain broad continuity of service, while micro-cells provide capacity in smaller high-demand areas. The two layers need coordinated channel assignments and handoff rules so that users can move between them without unnecessary connection interruption.

The preprocessing work should be completed before the final frequency groups are assigned. Traffic data identifies where capacity is needed; propagation measurements show where signals can reach; the hexagonal grid provides a starting geometry; and site verification confirms that equipment can be installed. The operator bears these planning costs as part of the project rather than treating them as optional activities.

Each strategy should be judged against the same service targets and the same available spectrum. The comparison should include SIR, blocking, channel allocation, site count and implementation complexity. This prevents a strategy from appearing better only because it was evaluated under more favourable assumptions than the alternatives.

The strategy selection should be reviewed whenever the township layout changes. A new academic building or hostel can create a new traffic hotspot and alter propagation paths. The frequency plan should therefore be maintained as a living engineering document, with each site and channel change recorded and checked against neighbouring cells.

An effective plan also needs coordination between radio planners and the installation team. The planned antenna height, azimuth, tilt and site location must match what is installed. If the physical configuration differs, the predicted interference and coverage may no longer be valid. Post-installation verification is therefore part of the frequency-planning process.

TASK 2 – Identify the Governing Design Parameter (2 Marks)

Fixed reuse: the governing parameter is cluster size N. Increasing N increases $D/R = \sqrt{(3N)}$ and improves SIR, but reduces channels per cell from 336/N. Sectorisation: the main parameters are sector angle and the number of dominant co-channel interferers i_0; narrower sectors generally reduce interference but increase antenna and planning complexity. Dynamic borrowing: the governing parameters are the number of free eligible channels and the blocking/interference threshold. Cell splitting: radius R and cells per area govern spatial capacity and site count.

As demand grows from 1,000 to 100,000+ subscribers, the design must add spatial reuse by splitting overloaded cells, improve sectorisation where interference is dominant, and use dynamic assignment to reduce avoidable blocking. N cannot simply be made very small if the 18 dB SIR target is violated.

Engineering Analysis and Application

The cluster size N is the central parameter for fixed reuse because it appears in both the channel allocation and the reuse-distance calculation. The number of channels per cell is approximately S/N, while the reuse ratio is $Q = \sqrt{(3N)}$. If N is increased, co-channel cells are separated by a larger multiple of the cell radius, but each cell receives fewer channels. This direct relationship makes N the most useful first parameter for comparing fixed reuse alternatives.

For sectorisation, the sector angle and the antenna pattern are important governing quantities. The sector angle determines the nominal coverage region, while the antenna pattern determines how much energy is received from different directions. The number of effective interferers, i_eff, is a useful simplified measure because the sector antenna can reduce the contribution from interferers outside its main beam. In a detailed model, i_eff should be replaced by the sum of individual interference powers weighted by antenna gain.

For dynamic assignment, the key quantity is the number of free channels that are eligible under the interference constraint. A channel may be unused in a neighbouring cell but still be unsuitable for borrowing if it would create strong co-channel interference. The controller must therefore consider both channel availability and radio compatibility. The traffic offered to the requesting cell is also important because it determines how frequently borrowing is required.

For cell splitting, cell radius R is the main geometric parameter. Cell area varies as R^2, so reducing the radius increases the number of cells that can cover the same township area. If R is halved, the area of each ideal hexagonal cell becomes one quarter of its previous area. The network can therefore create more local reuse opportunities, although the additional sites and handoffs increase cost and operational complexity.

As the subscriber population grows, the governing parameters should not all be changed in the same way. A larger N may improve SIR but reduce per-cell channels, so it is not always the correct response to rising traffic. Sectorisation is appropriate when interference is the limiting factor. Borrowing is appropriate when traffic is uneven and spare channels exist nearby. Cell splitting is appropriate when a location remains overloaded even after channel and antenna optimisation.

The analysis should report the parameter, the equation in which it appears, the effect of increasing or decreasing it, and the practical cost of changing it. This makes the comparison meaningful. A parameter is useful for design only when its impact can be related to measurable outcomes such as SIR, blocking probability, channels per cell, site count or handoff activity.

Further design considerations

The governing parameter should be chosen because it controls a major outcome in the design, not simply because it is easy to calculate. For fixed reuse, N controls the number of channels assigned to a cell and the distance to co-channel cells. For sectorisation, the antenna pattern

controls the angular suppression of interference. For dynamic assignment, the eligible free-channel pool controls whether a request can be served without violating interference limits.

Some parameters are linked. For example, Q is calculated from N, and D is calculated from Q and R. It is useful to distinguish a direct design choice from a derived quantity. The planner chooses a cluster size and cell radius; the reuse ratio and reuse distance then follow from the geometry. This distinction makes the design process easier to explain and prevents treating related quantities as independent controls.

Traffic growth does not necessarily mean that N should be increased. If traffic is the main problem, a larger cluster can reduce channels per cell and make blocking worse. The planner should first determine whether the observed limitation is interference, insufficient channel resources, poor coverage or a combination. The correct governing parameter depends on the limiting condition.

A sensitivity table can make the governing parameters easier to compare. Each row can show the parameter, the equation in which it appears, the effect of increasing it, the benefit, the disadvantage and the strategy affected. Such a table provides a direct link between the mathematical analysis and the practical decision for the township.

The final report should explain how each parameter is measured or controlled in practice. Cluster size is set in the frequency plan; sector angle and antenna gain come from antenna specifications and configuration; eligible channel count comes from network occupancy and interference checks; and cell radius is estimated from coverage planning and then verified by field measurements.

Performance and Planning Considerations

A parameter should be evaluated over the full range of values specified in the assignment. For example, the cluster-size test should include N = 3, 4, 7 and 12 rather than only the selected value. This reveals the direction of change and makes it possible to justify why N = 7 is selected as a compromise under the chosen SIR target.

The governing parameter should also be linked to the measurement used to verify it. SIR can be measured or estimated from received signal and interference levels; channel occupancy can be obtained from network counters; and cell radius can be checked through coverage measurements. This makes the analysis practical rather than purely symbolic.

When several parameters interact, the report should state which one is the main design control and which ones are supporting quantities. For example, cluster size N is a direct reuse choice, while Q and D are calculated from N and R. This clear distinction improves the technical explanation and avoids repeating the same relationship under different names.

The final parameter choice should be revisited after the traffic and propagation surveys. If the measured environment differs from the initial assumption, the selected cluster or sector plan may need adjustment. The mathematical relationship provides a starting point, while field measurements determine whether the selected values are suitable.

Implementation and planning notes

The relationship between cluster size and channel allocation can be presented as a simple trade-off. When N increases, the number of channels per cell decreases, but the reuse distance ratio

increases. The design must choose a value that satisfies the SIR requirement while retaining enough channels for the offered traffic.

For a sectorised cell, antenna gain is not identical in every direction. The serving sector should have sufficient gain toward its intended coverage area, while the gain toward dominant interferers should be lower where possible. The antenna pattern is therefore a practical governing input that should be taken from the manufacturer's data and checked in the field.

Dynamic assignment is governed by eligibility, not merely by whether a channel is idle. A channel should be selected only after checking its interference relationship with the requesting cell and the surrounding cells. This makes the interference threshold and the current set of free channels essential inputs to the allocation decision.

Cell radius is a direct control for the geographic scale of a cell. A smaller radius can improve local capacity by increasing the number of cells in a given area, but it also raises the number of sites and the frequency of handoffs. The radius should be reduced only where the measured traffic or coverage condition justifies the extra infrastructure.

The report should explain the effect of each parameter using both equations and plain language. A reader should be able to understand what happens when the parameter is increased, why the change occurs and what disadvantage accompanies the improvement. This is the purpose of identifying a governing design parameter.

At each subscriber milestone, the operator should review the governing parameter for the dominant problem. The correct response at 1,000 users may be different from the response at 100,000 users because the network may move from a coverage-limited condition to a capacity-limited condition.

The governing parameter is not necessarily the only parameter that matters. It is the most useful quantity for explaining the principal behaviour of a strategy. Other variables should still be recorded so that the analysis remains complete and the selected value can be reproduced.

A clear design report should show the selected parameter, its chosen value, the reason for selection and the measurement that will confirm it in the deployed network. This connects the theoretical discussion to the practical operation of the township system.

TASK 3 – Derive the Capacity and Interference Functions Mathematically (6 Marks)

1. Hexagonal geometry and reuse

For a regular hexagonal cell of centre-to-vertex radius R,

the area is $A_{cell} = (3\sqrt{3}/2)R^2 \approx 2.598R^2$ km². The approximate number of cells covering area A_T is $C = \text{ceil}(A_T/A_{cell})$. For R = 0.5 km,

$A_{cell} = 2.598(0.5^2) = 0.6495$ km² and $C = \text{ceil}(12/0.6495) = 19$ cells.

A valid cluster size is $N = i^2 + ij + j^2$. The co-channel reuse distance is $D = R\sqrt{(3N)}$, so reuse ratio $Q = D/R = \sqrt{(3N)}$.

For N = 7, $Q = \sqrt{21} = 4.583$ and $D = 0.5(4.583) = 2.291$ km.

2. Channels and Erlang-B capacity

Channels per cell $m = S/N = 336/7 = 48$.

Erlang-B blocking is B(m,A) = [A^m/m!] / Σ(k=0 to m)[A^k/k!],
where A is offered traffic in Erlangs. It can be calculated stably using $B_0 = 1$ and B_j = (A B_(j−1))/(j + A B_(j−1)). For m = 48, B(48,38) ≈ 1.80%, while B(48,39) is above 2%; therefore approximately 38 Erlangs per cell is the planning limit at 2% blocking.
Total offered traffic is A_total = U × 0.025 Erlang.
Thus 1,000 users offer 25 Erlangs; 10,000 offer 250; 20,000 offer 500; and 100,000 offer 2,500 Erlangs. A rough capacity bound is A_capacity ≈ C × 38 Erlangs, subject to traffic distribution and interference constraints.

3. Co-channel SIR

For equal-power base stations and path-loss exponent n, SIR = S/(ΣI_j). Under the six-equidistant first-tier interferer approximation, each interferer is at distance D and the desired mobile is approximated at distance R, giving SIR ≈ (D/R)^n/6 = (3N)^(n/2)/6. For N = 7 and n = 4: SIR = $21^2/6$ = 73.5, or $10\log_{10}(73.5)$ = 18.66 dB. This agrees with the assignment's sanity-check value of about 18.7 dB. For n = 2.5, the same approximation gives about 10.7 dB, illustrating that open environments can have worse co-channel SIR under this simplified model. For sectoring, replace six equal interferers with the smaller set of interferers that fall within the serving sector's antenna pattern. A simplified expression is SIR ≈ (D/R)^n/i_eff, where i_eff is the effective number of interferers after antenna attenuation. Real analysis should weight each interferer by its antenna gain and actual distance.

4. Cost–capacity relationship

Let c_site be the installed cost per site, C the number of sites, c_survey the survey cost, and c_op the operating cost per site per planning period. Then C_plan = c_survey + C(c_site + c_op). A useful cost-efficiency measure is cost per carried Erlang: J = C_plan / A_carried, where A_carried = A_offered(1 − B) for a uniform model. Fixed reuse has lower complexity but fixed channel groups; sectorisation adds antenna cost; dynamic borrowing adds coordination cost; cell splitting adds site and backhaul costs but increases spatial capacity.
Illustrative site planning: if c_site = ₹8,00,000, c_survey = ₹2,00,000 and C = 19, initial capital planning cost is ₹1.54 crore (₹2 lakh + 19 × ₹8 lakh), excluding spectrum, backhaul and recurring operations. These are assumptions for comparison, not quoted prices.

Design Principles and Practical Application

The hexagonal area equation follows by dividing a regular hexagon into six equilateral triangles. Each triangle has side length R and area $(\sqrt{3}/4)R^2$. Multiplying by six gives A_cell = $6(\sqrt{3}/4)R^2$ = $(3\sqrt{3}/2)R^2$. The numerical factor is approximately 2.598. This area equation is useful because it links the selected cell radius to the approximate number of sites required to cover the 12 km² township.
The number of cells is estimated by dividing the township area by the area of one ideal hexagonal cell and rounding upward. Rounding upward is necessary because a fraction of a cell cannot provide a complete coverage unit. The resulting number is only a first estimate. The

actual site count may be higher because the township boundary is not a perfect hexagon, cells overlap, indoor coverage is required and some candidate locations may be unavailable.

The cluster-size equation $N = i^2 + ij + j^2$ comes from the geometry of the hexagonal lattice. Moving i cell spacings along one grid direction and j spacings along a direction separated by 60 degrees gives a squared displacement proportional to $i^2 + ij + j^2$. The distance between co-channel cell centres is $D = R\sqrt{(3N)}$. Dividing by R gives $Q = D/R = \sqrt{(3N)}$. The ratio is dimensionless and is independent of the selected radius in this ideal model.

The traffic calculation begins with the offered traffic per subscriber. Since each subscriber offers 0.025 Erlang during the busy hour, the total is A_total = 0.025U. For a uniform first estimate, the traffic per cell is A_cell_traffic = A_total/C. This uniform division is convenient for a baseline calculation, but it should not be used to hide local congestion. A hostel cell may carry more traffic than a cell covering an open road, even when both cells have the same number of channels.

Erlang-B is a loss-system model. It assumes that a request is blocked when all channels are occupied and that blocked requests do not wait in a queue. The recursive formula is useful because it avoids large factorial values. Starting with $B_0 = 1$, each additional channel is included using B_j = A B_(j−1)/(j + A B_(j−1)). The final value after m iterations is the blocking probability for m channels and offered traffic A.

The simplified SIR expression is obtained by comparing the received desired signal with the sum of co-channel interference. If received power is proportional to distance raised to −n, the desired signal at distance R is proportional to R^(−n). If six first-tier interferers are each approximated to be at distance D, their combined interference is proportional to 6D^(−n). Dividing the two terms gives SIR ≈ (D/R)^n/6. Substituting $Q = \sqrt{(3N)}$ gives SIR ≈ (3N)^(n/2)/6.

Sectorised SIR should be evaluated using the antenna gain in the direction of each interfering base station. The desired signal is multiplied by the serving-sector gain, and each interference term is multiplied by the corresponding interferer gain toward the mobile. The simple i_eff expression is only a summary of this process. A more accurate calculation sums all individual interference powers, including side-lobe contributions, rather than assuming that all interferers outside the main beam disappear.

The cost–capacity relation should include both planning and recurring expenditure. Survey cost is incurred to estimate demand and propagation; site cost covers installation; operating cost covers power, maintenance, backhaul and site rental. A simple ratio of planning cost to carried traffic can compare alternatives, but it should use the same accounting period and should include the same cost categories for every strategy. Otherwise, a low-cost option may appear better only because some expenses were omitted.

Further design considerations

The cell-count calculation is an approximation because the township boundary may cut through cells. A real network may require partial cells along the edge, overlap between sites and additional sites for indoor coverage. Therefore, the calculated value of 19 cells is best treated as an initial geometric estimate. A site plan should be drawn over a map and adjusted to cover the actual buildings, roads and outdoor areas.

The traffic capacity calculation also depends on the definition of the busy hour. The value 0.025 Erlang per subscriber is an average busy-hour contribution. If a large number of users make calls at the same time, the demand may be higher than the average. The stadium challenge demonstrates this difference because attendee traffic is concentrated in one place and occurs over a short period.

The Erlang-B result should be interpreted as a probability under the assumptions of the model. A blocking probability of 1.8 percent means that approximately 1.8 percent of offered call attempts are blocked in the model, not that 1.8 percent of all users are permanently without service. Actual outcomes depend on arrival patterns, holding times, handoff treatment and the availability of alternative cells.

When comparing cost and capacity, the carried traffic is lower than the offered traffic whenever blocking occurs. A simple approximation is carried traffic = offered traffic × (1 − blocking probability). This can be used to compare how much traffic is successfully carried under two alternatives, provided the same traffic and accounting assumptions are used. It should not be confused with the total number of subscribers served.

The cost model can be extended to include site acquisition, equipment, antennas, installation labour, backhaul, electricity, maintenance, site rent and planning surveys. If the comparison is over several years, recurring costs should be discounted or accumulated consistently. The assignment provides no vendor quotation, so illustrative cost values should be clearly identified and replaced with local quotations in a real project.

Implementation Considerations

The area calculation also helps explain why a smaller radius increases the number of cells. Since A_cell is proportional to R^2, the number of cells covering a fixed township area is inversely proportional to R^2. If the radius is reduced to one half, the cell area becomes one quarter, and approximately four times as many cells are required for the same geographic area.

The reuse-distance equation can be used to compare cluster choices without selecting a particular radius. For N = 7, Q is $\sqrt{21}$. Once R is selected, the physical reuse distance follows by multiplying R by Q. This separates the geometric effect of cluster size from the physical scale of the cell and makes the equation easier to apply to different cell-radius assumptions.

The traffic model should be applied consistently. Total offered traffic is first calculated from the subscriber count, then distributed among cells according to the assumed traffic map. Erlang-B is applied to each cell using its channel count and offered traffic. The aggregate blocking performance should not be calculated by simply averaging cell blocking values if the cells have very different traffic loads.

Interference and blocking are separate performance measures. Increasing the reuse distance may improve SIR, but it can reduce the number of channels available to each cell and increase blocking. Adding channels or sites may reduce blocking, but if the new layout is poorly planned it can increase interference. A complete design therefore evaluates both quantities together.

Implementation and planning notes

The area formula is based on an ideal regular hexagon, so it should be used consistently with the stated radius definition. Here R is the distance from the cell centre to a vertex. If a different

radius convention is used, the area equation changes. Stating the convention prevents confusion when comparing cell sizes or checking calculations.

The approximate cell count is a geometric estimate and does not guarantee continuous service at every point. In a practical design, overlapping coverage may be required to support handoff and improve reliability. Boundary cells may also extend outside the township. These effects should be considered when the initial site count is converted into an actual deployment map.

The reuse-distance relation is useful for explaining how the same frequency group can be repeated. Co-channel sites must be separated enough to keep interference within the target. The physical distance D depends on both N and R, so changing the cluster size or cell radius changes the separation between co-channel sites.

The Erlang-B calculation should use the number of traffic channels available to the cell, not the total system channel pool. For the selected N = 7 cluster, the nominal allocation is 48 channels per cell. The traffic offered to each cell must then be calculated using the local subscriber distribution and the average traffic assumption.

Cost per carried Erlang can be used as a comparative measure when two plans carry different amounts of traffic. However, the comparison should include the same categories of expenditure and the same time period. If one option includes site rental and maintenance while another does not, the ratio will not be meaningful.

The equations in this task provide the mathematical basis for later simulation. The simulator should use the same definitions of radius, cluster size, channel count, traffic and SIR. If the code uses different definitions, the theoretical and simulated results cannot be compared fairly.

The mathematical model should be applied consistently from the beginning of the calculation to the final recommendation. The same radius convention, channel-pool definition, traffic assumption and path-loss exponent should be used in all related equations. Consistency is essential when comparing the analytical result with the simulator.

The final capacity result should always be described as idealised when it depends on uniform traffic and identical cells. Actual deployment capacity is affected by local loading, interference, equipment availability and site constraints. These factors should be checked before the mathematical estimate is used for investment decisions.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour (4 Marks)

Best SIR occurs near the serving base station, where desired signal power is high, and where the mobile lies away from the strongest co-channel directions. Average SIR occurs at a representative mid-cell position with ordinary shadowing. Worst SIR occurs near a cell edge/corner, especially when several co-channel cells are geometrically close and shadowing weakens the desired signal.

Using a path-loss model $P \propto d^{(-n)}$, the desired-to-interference ratio changes with the mobile's serving distance and its distances to interferers. At the cell edge, desired power falls as $R^{(-n)}$, while one or more interferers may be relatively close; therefore SIR decreases. With N = 7, n = 4 and six equidistant interferers, the reference SIR is 18.66 dB. If the effective interferer count doubles from 6 to 12 while distances stay fixed, SIR halves: $10\log_{10}(73.5/2) = 15.65$ dB.

Off-peak traffic normally has low blocking because offered traffic A is small. Average-hour blocking is moderate. Busy-hour or event-surge blocking is highest because A approaches or exceeds the Erlang-B limit. With 48 channels, A = 38 Erlangs gives approximately 1.80% blocking; at A = 40 Erlangs, blocking is higher than 2%. Shadowing can create local dead spots and raise outage probability. A larger path-loss exponent reduces distant interference more strongly, often improving reuse SIR, but dense buildings may also create obstruction and shadowing, so measurements are essential.

System-Level Explanation

The best-case radio condition occurs when the mobile is close to the serving base station and the desired path is not blocked. Since received power follows the approximate relation $P \propto d^{(-n)}$, a shorter serving distance produces a stronger desired signal. The mobile may also experience less interference if it is located in a direction where neighbouring co-channel sites are outside the strongest part of the serving-sector pattern.

The average case should represent the actual distribution of users rather than a single point at exactly half the cell radius. Users may be spread unevenly across buildings, roads and open spaces. A practical average can be estimated by sampling many mobile positions and calculating the SIR at each position. The mean, median and lower percentile should be reported because the mean alone can hide poor performance for users near the cell boundary.

The worst-case SIR is usually associated with a cell edge or corner where the serving base station is relatively distant and one or more co-channel interferers are comparatively close. The desired signal decreases with distance, while the interference sum may contain several strong contributions. The exact worst point depends on the geometry and on the position of neighbouring sites, so the six-equal-interferer equation should be treated as a reference approximation rather than a complete edge prediction.

Traffic conditions create a separate best-to-worst pattern for blocking. Off-peak traffic produces low channel occupancy and usually low blocking. Average traffic produces a moderate occupancy level. Busy-hour traffic brings the cell closer to its capacity limit, while an event surge can cause the offered traffic to exceed the planned capacity. Blocking can rise sharply near the capacity limit, so a small increase in offered traffic may cause a large change in service quality.

Shadowing changes the received power because buildings, trees and other obstacles obstruct the radio path. If the serving link is shadowed more strongly than the interfering links, SIR becomes worse. If an interfering link is shadowed, the interference contribution may decrease. The effect is therefore link-specific. A single average path-loss exponent cannot describe every location in a township with both open spaces and dense buildings.

The path-loss exponent changes the distance dependence of received power. In the simplified expression, a higher n causes distant interferers to decay more rapidly relative to the desired signal. However, dense built-up areas can also have deep local coverage holes due to building penetration and obstruction. The planning team must therefore evaluate both signal coverage and interference rather than concluding that a higher path-loss exponent always improves the network.

To present the cases clearly, the analysis should identify the mobile position, the serving distance, the approximate distances to dominant interferers, the path-loss exponent and the traffic load. The same assumptions should be used when comparing fixed reuse and sectorised layouts. This makes it possible to explain whether a change in performance comes from geometry, antenna directivity or traffic demand.

An operational network should include a margin below the maximum permitted blocking value. Designing exactly at 2% leaves little room for traffic variation, measurement error or temporary equipment faults. A reserve margin is particularly important in hostels and event areas where user activity can change quickly. The final margin should be chosen from measured traffic variation and not treated as an unsupported universal constant.

Further design considerations

A best-case result should not be used as the main measure of network quality. Users close to a base station may have a strong desired signal, but the service objective applies to the entire planned coverage area. Reporting only the best SIR can therefore make an incomplete network appear successful. Cell-edge and indoor conditions should be included in the analysis.

The average-case result should be weighted according to where users are expected to be located. If most subscribers are inside hostels, a uniform random distribution across open land will not represent the actual network. A traffic-weighted sample gives more importance to locations where users are concentrated and provides a more meaningful estimate of experienced performance.

The worst-case analysis should identify whether the limiting factor is a weak desired signal or a strong interferer. These conditions require different corrective actions. A weak desired signal may require a new site, a change in antenna tilt or improved indoor coverage. A strong interferer may require a different reuse assignment, sector orientation or power adjustment.

Time-of-day comparisons should distinguish ordinary busy-hour traffic from an event surge. The busy hour is a recurring planning condition, while a stadium event is an exceptional but predictable load. The event may create both traffic-channel congestion and control-channel access congestion. A design should therefore examine both the probability of a call being blocked and the ability of devices to access the network.

Shadowing should be included as a variation around the average path-loss prediction. A log-normal shadowing model is often used in radio planning, but the assignment does not provide a standard deviation. If such a model is added, the selected deviation must be stated as an additional simulation assumption rather than presented as supplied scenario data.

Operational Analysis

The best, average and worst cases should be defined before calculations are presented. The best case can use a mobile near the serving site, the average case can use representative sampled positions and the worst case can use a cell-edge or corner position. The report should state the assumed serving and interfering distances for each case so that the results can be reproduced.

An event surge can affect the network even if the average daily traffic is low. The stadium example concentrates a large number of users into a small area, and the resulting load may

exceed the local channel capacity. This demonstrates why traffic planning should include both recurring busy-hour conditions and known special events.

Radio quality should be evaluated separately from channel availability. A call may be blocked because every traffic channel is occupied even when the SIR is good. A call may also experience poor quality when channels are available but interference is high. The operator should identify which problem dominates before selecting a remedy.

The final case analysis should explain the likely operational response. Poor cell-edge SIR may lead to antenna adjustment or a new micro-site, while high busy-hour blocking may lead to additional channels through spatial reuse or cell splitting. The distinction makes the recommendations directly connected to the analysis.

Implementation and planning notes

The radio environment should be described by zones because the township contains open campus areas and dense built-up areas. A single path-loss exponent may not represent both environments. The selected value should be stated for each case, and the report should explain how the result changes when n moves from 2.5 toward 4.

The serving signal and the interference signals may be affected differently by obstacles. A building between the mobile and serving site may weaken the desired signal, while another building may block an interfering site. This is why the location of obstacles matters and why a simple distance-only model is an approximation.

Best-case SIR is useful as a reference for what the radio system can achieve under favourable conditions. Average-case SIR helps describe typical performance, while worst-case SIR identifies locations that may require optimisation. These three cases should be reported together so that a strong best-case value does not conceal a weak edge condition.

Blocking should be analysed using the offered traffic and the number of channels available. If a cell has 48 channels and offered traffic approaches the planning limit, the blocking probability rises. The event surge is an example of a condition in which local traffic may exceed the normal busy-hour design load.

The network should be designed with a margin because traffic and propagation measurements contain uncertainty. The margin should be based on the expected variation and the cost of adding capacity later. A design that exactly meets the threshold in an ideal calculation may fail when the real environment differs slightly from the model.

The final discussion should connect each observed problem to an appropriate action. Poor SIR suggests interference or coverage optimisation, while high blocking suggests additional traffic capacity. If both occur, a combined solution such as sectorisation and local cell splitting may be required.

The best, average and worst cases together provide a more complete description than a single SIR value. The report should identify where the network performs well, where typical users are likely to operate and where corrective action may be required. This is particularly important for a township containing both open and dense built-up areas.

The same approach should be used for blocking. Off-peak, average, busy-hour and event conditions should be compared using the offered traffic and available channels. This allows the

reader to understand how radio quality and traffic congestion affect the service in different ways.

TASK 5 – Scaling and Sensitivity Analysis (3 Marks)

For fixed N and channel pool S, channels per cell are approximately S/N, so per-cell capacity decreases approximately as 1/N. Reuse distance is $D/R = \sqrt{(3N)}$, and under the six-interferer approximation SIR $\propto$ N^(n/2). Cell count scales as $C \propto 1/R^2$. Halving R creates about four times as many cells over the same area.

Subscriber traffic grows linearly: 1,000 → 100,000 users increases offered traffic from 25 to 2,500 Erlangs, a factor of 100. At 38 Erlangs per cell, the idealised cell counts needed are ceil(25/38)=1, ceil(250/38)=7, ceil(500/38)=14 and ceil(2500/38)=66; practical coverage and geographic distribution can require more sites. At n = 2.5, the N = 7 simplified SIR is only about 10.7 dB, while at n = 4 it is 18.7 dB. Thus sectoring and measured propagation are important in open areas. Fixed reuse alone becomes costly or inflexible at high demand; dynamic borrowing helps short-term imbalances, while selective micro-cell splitting is preferable when overload is persistent.

Technical Background and Application

The first scaling relationship is channels per cell. With a fixed spectrum pool S, the nominal allocation is m = S/N. If the cluster size doubles, the nominal number of channels per cell is halved. This is why a very large cluster may not be suitable for a high-traffic cell even though it improves the co-channel reuse distance.

The second relationship is reuse distance. Since $Q = \sqrt{(3N)}$, the reuse ratio grows with the square root of cluster size. It does not double when N doubles. The SIR expression also depends on the path-loss exponent, so the effect of a change in N is different in open and dense environments. The assumed value of n should be selected from propagation measurements or a justified environmental model.

The third relationship is cell density. Hexagonal cell area is proportional to R^2, so the number of cells over a fixed area is approximately proportional to $1/R^2$. Halving R creates roughly four times as many cells in the same region. This increases opportunities for spatial reuse but also increases the number of base-station locations, backhaul connections and handoff boundaries.

Subscriber traffic grows linearly under the given per-subscriber traffic assumption. The increase from 1,000 to 100,000 subscribers is a hundredfold increase in total offered traffic. A network that is initially lightly loaded must therefore evolve. The evolution should be based on the local traffic distribution because the same total number of subscribers can create very different loads depending on where people gather.

The practical break-even point for sectorisation occurs when the improvement in interference and carried traffic justifies the additional antenna, radio and installation cost. The break-even point for dynamic assignment occurs when enough unevenness exists for borrowing to reduce blocking, while neighbouring cells still have spare eligible channels. The break-even point for cell splitting occurs when a cell remains overloaded for long enough that the cost of a new site is justified.

No exact financial break-even value can be obtained from the assignment data because equipment prices, site rental, backhaul charges and maintenance rates are not specified. The correct approach is to define the cost model, obtain local quotations and compare the cost per carried Erlang over a consistent planning period. This avoids presenting an unsupported financial result as a measured fact.

The sensitivity study should vary N, n, R and traffic demand one at a time before evaluating combined changes. Changing several parameters simultaneously makes it difficult to explain which design decision caused an improvement or deterioration. A final combined scenario can then be used to test the selected hybrid plan under realistic township conditions.

Further design considerations

A useful sensitivity analysis changes one input at a time while keeping the others fixed. For example, compare N = 3, 4, 7 and 12 at the same cell radius and path-loss exponent. Then compare n = 2.5, 3 and 4 using the same cluster size. This separates the effect of cluster size from the effect of the propagation environment.

The number of cells required for coverage and the number of cells required for traffic are not always the same. Coverage determines where radio sites are needed to reach the whole township, while capacity determines where additional resources are needed to carry busy-hour traffic. A network may have enough aggregate traffic capacity but still contain a coverage hole, or it may cover the entire area but have overloaded cells.

The break-even point should be identified separately for each strategy. Sectorisation is justified when directional interference limits performance; borrowing is useful when neighbouring cells have spare eligible channels; and cell splitting is useful when a local traffic hotspot persists. Treating all three as interchangeable would hide the reason for selecting a particular upgrade.

Subscriber growth may be accompanied by changes in user behaviour. A larger student population may increase traffic in hostels and academic blocks more than in open campus areas. Therefore, the per-subscriber traffic assumption should be reviewed when new usage measurements become available. The assignment's 0.025 Erlang value is retained for the required calculations, but future planning should use observed busy-hour data.

The network should be expanded in stages. After each capacity upgrade, the operator should collect new measurements and update the traffic map. This avoids installing too many sites too early and helps detect whether a problem has moved to a neighbouring cell. Staged expansion also makes it easier to compare the cost of each upgrade with the additional traffic carried.

Design Evaluation

The subscriber stages in the assignment provide a clear way to examine growth. The total offered traffic is 25, 250, 500 and 2,500 Erlangs at the four subscriber levels. The design should be checked at every stage because a strategy that works at 1,000 subscribers may not provide enough capacity at 100,000 subscribers.

Environmental sensitivity should be assessed using the full path-loss range in the scenario. The open-campus value near 2.5 and the dense built-up value near 4 produce different SIR estimates under the same cluster size. The report should explain that these are model values and that the actual environment may vary across the township.

A design may stop being practical when its cost, site count or handoff load becomes too high. The assignment does not provide exact vendor or operating costs, so this point should be described through the cost model and the measured traffic trend rather than an invented monetary threshold.

The recommended approach is to use the simplest strategy that meets the measured requirement, then add complexity only where it provides a demonstrated benefit. This avoids overbuilding the network and supports a staged investment plan as subscriber demand increases.

Implementation and planning notes

The number of cells required for coverage depends on the selected radius, while the number required for capacity depends on traffic and the capacity per cell. These two estimates should be compared. If the capacity estimate is larger, additional cells may be needed even though the original coverage grid already reaches the whole township.

The subscriber growth calculation is linear because the assignment gives a constant average busy-hour traffic per subscriber. If the average traffic changes, the total offered traffic changes in direct proportion. The assumption should therefore be reviewed if usage patterns change or if the network begins carrying a different mix of services.

The path-loss exponent sensitivity test demonstrates that the propagation environment affects the interference calculation. The same cluster size may satisfy the SIR target under one assumed environment and fail under another. This is why the assignment asks for values from open campus to dense built-up conditions.

At low demand, a large number of micro-cells may be uneconomical because the additional sites would be lightly loaded. At higher demand, selective splitting becomes more attractive because the same infrastructure can carry additional traffic in the locations where it is needed. The timing of the upgrade should be based on observed growth.

The design should not use a single aggregate capacity number as the only decision measure. It should also examine local cell loading, radio quality, number of sites and mobility. A plan that has adequate total capacity may still fail in one busy hostel or stadium sector.

A staged network plan provides a practical way to respond to growth. The operator can begin with the baseline reuse plan, add sectorisation where required, enable borrowing for uneven traffic and install micro-cells in persistent hotspots. Each stage should be justified by measured performance.

The scaling analysis should be used as a decision aid rather than as a prediction of exact deployment cost. The equations show how the number of cells, channel allocation and reuse distance change when a parameter is varied. Site surveys and local quotations are needed to convert these relationships into a final investment plan.

The most suitable strategy may change as the township grows. A simple fixed plan may be enough at the initial stage, while a hybrid approach becomes more useful as demand becomes concentrated and the subscriber base increases. Periodic measurement ensures that the network evolves at the correct time.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting (3 Marks)

Let C_0 be the original number of cells, R_0 the original radius, and D_0 the offered traffic. One uniform split round reduces radius from R to R/2. Since cell area is proportional to R^2, each parent cell is replaced by approximately four smaller cells. Therefore C_k = 4C_(k−1), C_0 given, and hence C_k = $C_0$4^k. The radius is R_k = R_0/2^k.

If the same channel reuse pattern is maintained, the idealised area capacity scales with cell count: K_k = 4K_(k−1) = $K_0$4^k. This assumes each new cell can use the same per-cell channel allocation without violating interference constraints. With the assignment's power model, P_k = P_0/2^(nk), because P′ = P/2^n at each split. Smaller cells create more boundaries, so handoff attempts and mobility-management load increase.

For selective splitting, let f_k be the fraction of cells split in round k. Then C_k = C_(k−1)[1 + 3f_k], because each selected parent is replaced by four children, adding three cells. For a constant fraction f, C_k = C_0(1+3f)^k. The base case is k=0: C=C_0, R=R_0, P=P_0 and capacity K=K_0. Planning cost per round includes new sites, backhaul, power, permits and optimisation.

Detailed Design Explanation

The base case represents the network before any splitting takes place. It contains C_0 cells, each with initial radius R_0 and a defined channel plan. The offered traffic is D_0 or, when capacity is expressed in Erlangs, the initial supported capacity is K_0. The base case is important because every later recurrence must be traceable to the original network rather than starting from an arbitrary round.

In one uniform split, one parent cell is replaced by four child cells. The radius of each child is R/2. Because the hexagonal area is proportional to R^2, each child covers one quarter of the parent area. Four child cells therefore cover approximately the same total geographic area as the original parent cell, ignoring overlap and boundary effects.

The cell-count recurrence is C_k = 4C_(k−1). It follows because every cell in the previous round is replaced by four cells. Repeated application gives C_k = $C_0$4^k. This is an ideal uniform model. In an actual township, not every cell will need splitting and not every location will have a suitable site for four child cells.

The radius recurrence is R_k = R_(k−1)/2, which gives R_k = R_0/2^k after k rounds. The radius reduction creates shorter average serving distances and more spatial reuse opportunities. It also creates more boundaries between cells, so mobile devices may perform handoffs more frequently as they move through the network.

The assignment specifies the transmit-power relation $P' = P/2^n$ for one split. Repeating the same rule gives P_k = P_0/2^(nk). This is the recurrence implied by the exercise's model. In practical radio engineering, transmit power must be selected using coverage requirements, antenna gain, receiver sensitivity and interference measurements; the recurrence should not be applied as an automatic equipment setting.

For selective splitting, only a fraction f_k of the existing cells is selected for splitting. Each selected parent is replaced by four children, which adds three cells to the total. Therefore C_k = C_(k−1)(1 + 3f_k). If f_k is constant, C_k = C_0(1 + 3f)^k. The uniform case is recovered when f = 1, and no splitting occurs when f = 0.

The work in each round includes traffic analysis, site selection, installation, backhaul provision, power connection, channel reassignment and radio optimisation. The cost of a round depends on how many cells are split and the availability of suitable locations. Selective splitting can reduce unnecessary expenditure because low-demand cells remain unchanged.

The recurrence should be stopped when the required traffic capacity is reached, when blocking is within the target, or when a practical limit such as site availability or cost is reached. It should also be checked that the new layout does not create unacceptable interference or excessive handoff signalling. Capacity growth alone is not a sufficient stopping criterion.

Further design considerations

The recurrence for uniform splitting assumes that every cell is treated in the same way at each round. This is useful for deriving a closed-form expression, but it is not always a realistic deployment strategy. Some parts of the township may remain lightly loaded while other parts experience rapid growth. The selective recurrence is therefore important for practical planning.

The cell-count recurrence measures the number of cells, but a project manager also needs to estimate the work required in a round. If every cell is split, the number of new sites grows rapidly. The work includes selecting locations, obtaining permissions, installing equipment, connecting backhaul and testing the radio configuration. The number of sites is a useful first indicator of this workload.

The capacity recurrence assumes that every child cell can carry the same traffic as the original cell. This is an idealisation. The actual capacity may be limited by the available channel groups, interference from nearby cells, backhaul throughput or local coverage. The simulator should test these conditions before the theoretical capacity multiplier is used in a deployment decision.

The power recurrence should be connected to the assignment's stated rule. It describes a mathematical reduction in transmit power after a radius reduction. In a real design, the operator should not simply reduce power without checking the coverage map. If power is reduced too far, users may experience weak signals inside buildings or near the edge of a micro-cell.

Handoff load is not necessarily proportional only to the number of cells. It also depends on the speed of users, their routes, the shape of coverage areas and the handoff thresholds. A dense grid near a road may generate many handoffs for moving users, while a dense grid inside a stationary hostel population may generate fewer handoffs per user. Mobility assumptions should therefore accompany the recurrence analysis.

Engineering Discussion

The recurrence should state clearly whether it describes all cells or only overloaded cells. Uniform splitting replaces every cell at each round and produces the multiplier of four. Selective splitting replaces only a fraction of cells and produces a smaller multiplier. Mixing these two assumptions in one equation would lead to an incorrect estimate of site growth.

The base case should include the initial number of cells and the initial radius. If the original coverage estimate is 19 cells, then C_0 can be set to 19 for an illustrative uniform model. The actual value used should match the network stage being analysed and should not be changed between the recurrence and its numerical example.

The number of new sites in a selective round is three times the number of parent cells selected for splitting. This follows because each selected parent is replaced by four children, so the net increase is four minus one. This gives the factor 1 + 3f in the selective recurrence.
The recurrence is a planning model for repeated network densification. Its results should be checked against available locations and the actual radio plan after every round. A mathematical recurrence cannot guarantee that the required sites, backhaul or permits will be available.

Implementation and planning notes

The recurrence should identify the parent cell and the child cells created by one split. The parent is removed from the active layout and replaced by four smaller cells. This explains why the net increase in the number of cells is three rather than four for each parent selected.
Cell splitting changes the network geometry and therefore requires a new channel and neighbour plan. The new cells may be closer to co-channel sites than the original parent was. The operator must verify the reuse distance and interference after each split instead of assuming that the old plan remains valid automatically.
The offered traffic does not necessarily divide equally among the four child cells. The distribution depends on where users are located within the parent cell. A hostel or lecture building may be concentrated in only one part of the parent area, so the child cells may have very different traffic loads.
The cost of a split round depends on the number of parent cells selected and the physical difficulty of installing each new site. Selective splitting is therefore more realistic for a township with uneven traffic. The fraction f should be based on the proportion of cells that actually exceed the chosen planning threshold.
The recurrence provides a compact way to estimate the scale of the expansion. It should be accompanied by a description of the assumptions about capacity per child cell, transmit power and handoff behaviour. This ensures that the mathematical result is not interpreted as a complete deployment design.
After each round, the operator should recalculate traffic per cell and check whether the blocking target is met. If the overloaded cells are still above the target, another selective round may be required. If capacity is adequate, further splitting should be avoided unless coverage or reliability requires it.
The recurrence is useful because it provides a clear relationship between one planning round and the next. It identifies the expected change in cell count, radius and capacity. The model should always be accompanied by its base case and assumptions so that the result can be interpreted correctly.
Selective splitting is particularly relevant to the township because demand is unlikely to be equal in every cell. Applying the split only to overloaded cells can reduce unnecessary infrastructure and keep the network expansion aligned with actual traffic.

TASK 7 – Solve the Recurrence and Interpret the Result (4 Marks)

For uniform splitting, C_k = 4C_(k−1). Substitution gives $C_1=4C_0$, $C_2=4(4C_0)=4^2C_0$, and by induction C_k=$C_0$4^k. If original capacity is K_0 and demand is D, require $K_0$4^k ≥ D. Dividing

by K_0 gives $4^k \geq D/K_0$; taking log base 4 gives $k \geq \log_4(D/K_0)$. The minimum integer number of rounds is $k = \max(0, \text{ceil}(\log_4(D/K_0)))$.
Example: if the current layout carries 700 Erlangs and forecast demand is 2,500 Erlangs, $k = \text{ceil}(\log_4(2500/700)) = 1$ round. The idealised capacity becomes 2,800 Erlangs. Sites become four times the original count and radius halves. This is a capacity approximation; channel availability, interference, and nonuniform traffic must be checked.
For selective splitting with constant split fraction f, capacity multiplier per round is approximately (1+3f), so $k = \max(0, \text{ceil}(\log(D/K_0)/\log(1+3f)))$. If f=0, no capacity is added. More splitting increases site/backhaul cost and handoff rate; selective splitting avoids spending money in low-demand areas.

Technical Explanation and Application

Iteration is a suitable method for solving the uniform recurrence because each round is defined directly in terms of the previous round. Starting with C_0, the first round gives $C_1 = 4C_0$. Substituting again gives $C_2 = 4C_1 = 4^2C_0$. The third round gives $C_3 = 4C_2 = 4^3C_0$. The repeated pattern leads to $C_k = C_0 4^k$.
The same idealised multiplier is used for capacity when every new cell is assumed to provide the same per-cell capacity as the original cell. Thus $K_1 = 4K_0$, $K_2 = 4K_1 = 4^2K_0$ and $K_k = K_0 4^k$. This assumes that the channel allocation and interference conditions remain suitable after every split. If these assumptions fail, the actual capacity increase will be smaller.
To determine the number of rounds, the network must satisfy $K_0 4^k \geq D$, where D is the forecast demand. Dividing by K_0 gives $4^k \geq D/K_0$. Taking logarithms gives $k \geq \log_4(D/K_0)$. Since only whole splitting rounds are possible, the result is rounded upward. If the current capacity already exceeds demand, no splitting round is needed.
For selective splitting with constant fraction f, the capacity multiplier is approximated by $(1 + 3f)$. The corresponding recurrence gives $K_k = K_0(1 + 3f)^k$. The minimum round count is the ceiling of $\log(D/K_0)$ divided by $\log(1 + 3f)$, provided f is greater than zero and the demand exceeds the initial capacity. This gives a way to compare uniform and selective growth.
The closed-form result shows that uniform splitting grows rapidly. The cell count and idealised capacity are multiplied by four in each round, while radius is divided by two. This can solve a capacity problem quickly, but it can also produce a large number of sites. The cost may grow faster than the operator can justify if many cells are split even though their traffic is low.
The transmit-power recurrence in the assignment decreases power by a factor of 2^n per round. For n = 4, the modelled power after one split is $P_0/16$. This result should be interpreted as the mathematical rule specified for the assignment, not as a field recommendation. Actual power levels must be validated against coverage and interference measurements.
Smaller cells increase handoff frequency because a moving user crosses cell boundaries more often. The effect is stronger for users travelling along roads or moving between buildings. Handoff signalling consumes network resources and requires accurate neighbour lists. A practical split plan must therefore include mobility analysis and should avoid creating very small cells in locations where users move rapidly unless the capacity benefit is necessary.
Selective splitting is generally more economical when overload is concentrated in a limited number of cells. The operator should rank cells by busy-hour occupancy, blocking and traffic

growth, then split the most overloaded cells first. After each round, the neighbouring cells should be rechecked because traffic and interference may shift after the new sites become active.

Further design considerations

The logarithmic solution gives a minimum number of ideal splitting rounds. The ceiling operation is required because a fractional round cannot be implemented. For example, if the calculation produces 1.2 rounds, the design must complete two full rounds to meet or exceed the required capacity under the model. The resulting capacity may be higher than the exact demand, which should be considered when estimating cost.

If the initial capacity already meets demand, the number of additional rounds is zero. This condition should be stated explicitly because the logarithm of a ratio below one is negative, but a negative number of splitting rounds has no physical meaning. The planning expression therefore uses a lower bound of zero.

The selective formula becomes more difficult when the split fraction changes from one round to another. In that case, the capacity multiplier is the product of the round-specific multipliers. The cell count can be written as $C_k = C_0$ multiplied by $(1 + 3f_r)$ for each round r. This allows the split fraction to reflect changing demand or site availability over time.

The closed-form equations are planning tools rather than guarantees. If a child cell cannot be installed or if its radio resources are restricted by interference, the actual capacity increase will be smaller than fourfold. The final round count should therefore be checked using a realistic channel plan and local site constraints.

A practical expansion decision should compare the cost of one more splitting round with the expected benefit. If the demand forecast is uncertain, the operator may choose a smaller selective expansion first and review the measured traffic after deployment. This staged approach reduces the risk of paying for capacity that remains unused.

Practical Design Considerations

The solution method should be shown step by step so that the closed form can be verified. Substitution demonstrates the pattern, and mathematical induction can be used to confirm it for all non-negative integer rounds. The logarithmic expression then follows from the capacity requirement and the need to solve for k.

The selective formula should be interpreted carefully when no cells are selected. If f is zero, the multiplier is one and capacity does not grow, so the logarithmic denominator is zero. In that case, the network cannot reach a higher demand through splitting under the selected policy and another action is required.

The site count is a useful cost indicator, but it is not the full cost. A new site may require different civil work, power and backhaul depending on its location. A rooftop micro-cell may be less expensive than a new ground site, while a remote site may need additional connectivity. Cost estimates should reflect these differences where data is available.

The increase in handoff activity should be discussed with the capacity benefit. Smaller cells can improve local capacity but may create more handoff events for moving users. The final design should balance capacity, mobility and operating complexity rather than selecting the maximum number of splitting rounds.

Implementation and planning notes

The closed form C_k = C₀4^k makes the rapid growth in site count clear. For example, every additional uniform round multiplies the existing cell count by four. This can create a large deployment very quickly, which is why the method should be applied only when the expected traffic benefit justifies the cost.

The minimum-round formula is based on a capacity requirement. The forecast demand is compared with the capacity after k rounds, and the smallest whole number satisfying the inequality is selected. The ceiling operation ensures that the design does not stop at a fractional round that cannot be implemented.

The selective solution is useful when only part of the network is overloaded. If the split fraction is small, the capacity multiplier per round is smaller than four, but the operator also installs fewer sites. The total cost may therefore be lower even if more planning rounds are required.

The transmit-power expression should be presented as the assignment's mathematical assumption. Actual power control is a radio optimisation task and depends on the required coverage and interference environment. The report should not convert the recurrence directly into a hardware setting without validation.

The handoff impact should be considered in the same way as site cost. A denser layout may support more traffic, but moving users may cross cell boundaries more frequently. This can increase signalling and place additional work on the network controller.

The final interpretation should state the benefits and disadvantages of uniform and selective splitting. Uniform splitting is simple to model and gives a rapid capacity increase, while selective splitting better matches uneven demand and avoids unnecessary sites in lightly loaded areas.

The number of rounds should be interpreted together with the resulting site count. A mathematical solution that meets the traffic requirement may still be impractical if it requires too many sites or creates excessive handoff activity. The operator should compare the closed-form result with physical site availability and operating constraints.

The selective solution gives the operator more control over where investment is made. It may take additional rounds to achieve the same theoretical capacity, but it can avoid expanding low-demand areas. This makes it a useful option for a network with clearly identified hotspots.

TASK 8 – Implement and Test the Planning Strategies (4 Marks)

The following Python program is a compact reproducible baseline. It calculates valid cluster sizes, channels per cell, hexagonal cell count, Erlang-B blocking, and the six-interferer analytical SIR. It uses a fixed random seed for reproducibility. It is an analytical planning simulator; a full Monte-Carlo spatial simulator would additionally generate mobile coordinates, shadowing samples, antenna patterns and individual interferer distances.

```
import math
import random

random.seed(7)
```

```
def erlang_b(channels, traffic):
    b = 1.0
    for m in range(1, channels + 1):
        b = traffic * b / (m + traffic * b)
    return b

def valid_clusters(limit=20):
    values = set()
    for i in range(limit + 1):
        for j in range(limit + 1):
            n = i*i + i*j + j*j
            if n > 0:
                values.add(n)
    return sorted(values)

def hex_area(radius):
    return (3 * math.sqrt(3) / 2) * radius * radius

def sir_db(n_cluster, path_loss=4.0, interferers=6):
    sir = (3 * n_cluster) ** (path_loss / 2) / interferers
    return 10 * math.log10(sir)

S = 336
radius = 0.5
area = 12
users = [1000, 10000, 20000, 100000]

print("N, channels/cell, SIR dB, cells, traffic, blocking")
for N in [3, 4, 7, 12]:
    channels = S // N
    cells = math.ceil(area / hex_area(radius))
    for U in users:
        traffic = U * 0.025 / cells
        blocking = erlang_b(channels, traffic)
        print(N, channels, round(sir_db(N), 2), cells,
              round(traffic, 2), round(blocking * 100, 2))

print("Valid cluster sizes:", valid_clusters())
```

Test cases required by the assignment are N = 3, 4, 7, 12; n = 2.5, 3, 4; sectorised and unsectorised layouts; and subscriber levels 1,000, 10,000, 20,000 and 100,000. The known check N=7, n=4 should produce about 18.66 dB. N=3 gives a lower SIR under the same approximation and should be flagged if it does not meet the selected 18 dB target.

Engineering Analysis and Application

The supplied program is a compact analytical baseline. Its functions have separate responsibilities: the cluster generator creates valid hexagonal cluster sizes, the area function calculates the ideal cell area, the Erlang-B function estimates blocking, and the SIR function calculates the six-interferer reference. Keeping these functions separate makes the program easier to test and makes it possible to replace one model without rewriting the complete program.

The random seed is set to a fixed value to support reproducibility. A fixed seed is important when random sampling is used because it allows two strategy runs to use comparable random sequences. The current compact code does not generate mobile coordinates or calculate individual interferer distances. Therefore its SIR output is an analytical reference rather than a full Monte-Carlo spatial distribution.

The cluster-size generator uses $N = i^2 + ij + j^2$. It stores values in a set so that duplicate cluster sizes are not repeated. The output is sorted to make the candidate list easy to inspect. The upper limit for i and j controls how many candidate values are explored; it should be increased if a larger cluster size is required for a different planning case.

The Erlang-B routine starts with a blocking value of one and updates it once for each channel. This recursive calculation is numerically stable for the channel counts used in the assignment. It should be tested with known input values and compared with a trusted Erlang-B table or an independent calculation. A result should not be accepted merely because the program runs without an error.

The SIR routine first calculates the linear ratio and then converts it to decibels using $10\log_{10}$. The distinction between linear and decibel values is important. Interference powers must be added in linear units; adding decibel values directly would produce an incorrect result. The program's reference check for $N = 7$ and $n = 4$ should be approximately 18.66 dB.

To extend the baseline into a spatial Monte-Carlo simulator, generate mobile coordinates inside each hexagonal cell. For each mobile, calculate its distance to the serving base station and to each relevant co-channel interferer. Convert distances into received powers using the selected path-loss exponent, apply antenna gains where sectorisation is enabled, sum interference in linear units and calculate the SIR for that sample.

The spatial simulator should also allow a shadowing term and a consistent random seed. The same mobile coordinates and shadowing samples should be used when comparing strategies, where practical. This paired comparison reduces the effect of random sampling differences and makes it easier to attribute changes in SIR to the planning strategy.

Testing should cover all specified cluster sizes, path-loss exponents, subscriber levels and sector configurations. Each run should record the input parameters and output values. The code should be checked for boundary cases, including a small traffic load, a high traffic load, a cluster size that does not divide the channel pool evenly and a cell radius that produces a non-integer number of geometric cells.

Further design considerations

Code validation should be carried out at the function level before running the full parameter matrix. The area function can be checked using a known radius and the expected hexagonal

area. The cluster generator can be checked to confirm that each output can be expressed as $i^2 + ij + j^2$. The Erlang-B function can be checked against a known table value, and the SIR function can be checked against the N = 7 reference.

The program should print the input assumptions before the results. This includes the total channel pool, cell radius, township area, subscriber values, path-loss exponent and blocking target. Printing these values reduces the chance that a result is copied into the report without the conditions under which it was calculated.

Integer division in channel allocation requires attention. If the channel pool is not evenly divisible by the cluster size, some groups may receive one more channel than others. The simple floor operation is a conservative estimate, but an actual frequency plan should distribute the remainder and verify that the allocation does not create an interference conflict.

The program should also distinguish a cell count used for geographic coverage from a cell count used for capacity. The current baseline uses the geometric cell count and divides total traffic evenly across it. A more complete simulator should calculate traffic in each cell separately, especially when the stadium, hostels or lecture buildings create localised demand.

Test results should be saved in a structured format such as CSV so they can be checked and plotted later. Each record should include the strategy name, cluster size, radius, path-loss exponent, subscriber count, channel count, offered traffic, blocking probability, SIR result and random seed. This makes the experiment repeatable and supports a transparent comparison.

Performance and Planning Considerations

The code should include comments or documentation in the submitted program so that another student can understand the purpose of each function. The assignment answer may show the main code and then explain its inputs and outputs. Function names should be descriptive, and the same units should be used throughout the program.

A useful validation step is to test the program with a very small offered traffic value and confirm that blocking is close to zero. A high offered traffic value should produce a larger blocking probability. These directional checks do not replace comparison with known values, but they help identify basic programming mistakes.

The spatial Monte-Carlo extension should report the number of sampled mobile positions. A small sample may produce unstable percentiles, while a larger sample gives a more reliable estimate at the cost of longer runtime. The selected sample count should be recorded so that results can be reproduced.

The program should not claim to simulate sectorisation unless it actually applies sector antenna gains or a clearly defined effective-interferer model. Similarly, it should not claim to implement dynamic borrowing unless it tracks channel availability and checks the interference rule. The report should describe the implemented features accurately.

Implementation and planning notes

The implementation should be divided into functions so that each part can be tested separately. A cluster generator, hexagonal area calculator, Erlang-B routine and SIR routine make the baseline program easier to understand. Additional functions can later be added for mobile positions, antenna gain and individual interference links.

The program output should be labelled as analytical or simulated. The current reference calculations are analytical. A true Monte-Carlo result requires random or measured mobile locations and a calculation of the desired and interfering signal at every location. This distinction should be maintained throughout the report.

The same random seed should be used for comparable strategy runs. When the mobile positions are identical, the difference in results is more directly related to the planning method. If different seeds are used, some variation may be caused by the random sample rather than the strategy itself.

The test matrix should include the four cluster sizes and three path-loss exponents specified by the assignment. It should also include both sectorised and unsectorised configurations. Each combination should be run at the four subscriber levels so that performance can be compared as demand grows.

Errors should be checked before results are copied into the report. The program should be run from a clean state, and the output should be saved. If a result changes after editing the code, the report should be updated so that it matches the final tested version.

The implementation section should explain what the program calculates and what it does not calculate. A compact analytical model is acceptable as a baseline, but the answer should not claim that it produces spatial percentiles or outage values unless those calculations are actually implemented.

The program should be accompanied by a short description of the execution environment and the input values. If the code is modified, the final output should be regenerated and checked. This helps ensure that the submitted code and the numerical results in the report refer to the same version.

Testing should include both expected values and unusual input conditions. The known SIR case and Erlang-B reference provide useful checks, while low-traffic and high-traffic cases test the direction of the blocking calculation. A reproducible program strengthens the credibility of the design analysis.

TASK 9 – Record and Present Experimental Performance (3 Marks)

Record, for every strategy and each traffic level: mean SIR, 5th-percentile SIR, outage probability (fraction below 18 dB), Erlang-B blocking probability, channels per cell, required sites and execution time. Keep the same random seed and mobile positions for fair comparisons. Suggested results table: Strategy | N | sector count | path-loss n | mean SIR (dB) | 5th-percentile SIR (dB) | outage (%) | blocking (%) | sites | runtime (s). Plot (1) SIR cumulative distribution functions, (2) blocking probability versus offered traffic, and (3) ideal capacity versus N. Label units, include assumptions and use identical axes when comparing strategies.

For the analytical reference with N=7 and n=4, SIR is 18.66 dB. With 48 channels, Erlang-B blocking is approximately 1.80% at 38 Erlangs per cell. These are reference values for validating the implementation, not claimed outputs from a full spatial Monte-Carlo experiment.

Design Principles and Practical Application

The mean SIR gives a broad indication of radio quality, but it does not show how many users experience poor conditions. The 5th-percentile SIR is useful because it describes a lower-tail result: approximately 95 percent of sampled positions have an SIR at or above that value. Outage probability measures the proportion of sampled positions whose SIR is below the selected threshold of 18 dB.

Blocking probability should be reported for every traffic level and every channel allocation. It is important to state whether traffic is divided uniformly across cells or assigned according to a measured heat map. A uniform traffic assumption can make the aggregate result appear acceptable even when one high-demand cell is overloaded.

Channels per cell should be reported alongside cluster size because a high SIR does not necessarily mean high traffic capacity. A large N may improve reuse separation while reducing the number of channels allocated to each cell. The number of sites should also be included because a design with many small cells may support more traffic but require more capital and operating expenditure.

Runtime is a useful measure when comparing algorithms with different levels of complexity. A fixed-reuse calculation may complete quickly, while a spatial Monte-Carlo model with many mobile samples and multiple interferer tiers may take longer. Runtime should be measured on the same computer and with the same sample count so that the comparison is fair.

The SIR cumulative distribution function should plot SIR on the horizontal axis and the cumulative proportion of mobile samples on the vertical axis. It allows the reader to compare the full distribution rather than only one average. The 18 dB target can be marked as a reference line so that the outage region is visible.

The blocking-versus-offered-traffic plot should show how the probability of a blocked call changes as the offered Erlangs increase. Each strategy should use a consistent channel count and traffic definition. A horizontal reference at 2 percent makes it possible to identify the approximate traffic limit for the selected grade of service.

Capacity-versus-N should show the trade-off between the reuse cluster size and channels available per cell, and should be interpreted together with SIR. A graph of channels per cell alone may favour small N, while a graph of SIR alone may favour large N. The best design must satisfy both requirements and should also be evaluated for site cost.

Every table and plot should identify whether its values are analytical, simulated or measured. If the program does not calculate a 5th percentile or an outage probability, those cells should be marked as not computed until a spatial simulation is added. This reporting discipline prevents unsupported experimental claims and makes the assignment reproducible.

Further design considerations

The results table should use consistent units and labels. SIR should be reported in decibels, blocking and outage should be reported as percentages, traffic should be reported in Erlangs, and site count should be an integer. The reader should not need to infer whether a value is linear, logarithmic or expressed as a percentage.

Mean and 5th-percentile SIR should be calculated from the same set of mobile samples. The outage percentage should be the proportion of those samples below 18 dB. If a strategy has no

spatial samples, the table should show that these metrics were not calculated rather than inserting a theoretical value in their place.
The blocking result should be linked to the actual number of channels available to each cell. If the channel allocation differs between strategies, that difference should be visible in the table. A design with a lower blocking probability may simply have more channels per cell, so the SIR and site-count columns are needed to interpret the result fairly.
Graphs should include a title, axis labels, units and a legend. The same axis limits should be used when comparing strategies. The 18 dB SIR threshold and 2 percent blocking target should be identified clearly so that the reader can see which alternatives meet the design requirements. The discussion below each graph should explain the trend rather than repeat every plotted number. For example, it should identify whether a change in cluster size improves SIR at the expense of channel count, or whether sectorisation improves the lower tail of the SIR distribution. This interpretation connects the simulator outputs to the original design objectives.

Implementation Considerations

The result table should include enough information to reproduce every row. At minimum, record the subscriber count, offered traffic, cluster size, path-loss exponent, radius, number of channels per cell and number of sites. For sectorised cases, also record the number of sectors and the antenna assumptions.
The results should be interpreted against the assignment's two main service targets: SIR of at least 18 dB and blocking of no more than 2 percent. A strategy that satisfies one target but fails the other should not be described as fully meeting the design requirement.
If multiple runs are performed with different random seeds, the report can show the average result and the range across runs. This helps distinguish a stable strategy improvement from a result that occurred because of one random sample. The number of runs and the seed policy should be stated.
The conclusion should refer to the data actually produced by the program. Analytical reference values can be included for validation, but they should be labelled as calculations. A table should not contain invented experimental values for metrics that the code does not calculate.

Implementation and planning notes

The experimental record should include the date of the run, code version, random seed and input parameters. This information makes it possible to reproduce the results later. It is especially important when the simulator is modified during development because the same output may not be produced by a different version of the code.
The SIR distribution should be presented using the same sample definition for all strategies. If one strategy uses a larger number of samples, its percentile estimate may be more stable. The sample count should therefore be consistent or any difference should be explained.
Blocking probability should be interpreted together with traffic and channel count. A low blocking value under light traffic does not prove that a strategy is better under high traffic. The full subscriber range is needed to show how the network behaves as the load increases.
The number of sites should be reported because densification may improve capacity but increase infrastructure cost. The runtime should be reported because more complex strategies may

require more computation. Together, these measures show both network performance and the effort required to evaluate it.
Tables should be concise enough to compare alternatives, while the accompanying discussion should explain the main trends. A graph should not be included only for appearance; it should answer a specific question such as how SIR changes with N or how blocking rises with offered traffic.
Where a metric is not available, the report should state that it was not computed in the current implementation. This is more accurate than filling the table with estimated or invented values. Additional simulation work can then be identified as a future improvement.
The presentation should make it possible to identify the best overall strategy without hiding disadvantages. A short discussion should state which option meets the SIR target, which meets the blocking target, how many sites are required and what additional complexity is introduced. Any limitations in the experimental data should be stated near the relevant table or graph. This makes the report transparent and prevents analytical estimates from being mistaken for measured field performance. It also identifies which additional experiments would improve the design study.

TASK 10 – Compare Theory with Simulation (4 Marks)

Theory and simulation should agree closely when the simulation uses six equal-distance interferers, equal transmit powers, no shadowing, and a mobile at the assumed reference position. The N=7, n=4 reference should be about 18.7 dB. Erlang-B should also match when the simulator uses the same number of channels and offered traffic.
Differences arise because real mobiles are not all at the same distance; the number and distances of interferers vary; shadowing and antenna patterns alter received powers; only a finite number of cell tiers may be simulated; and random sampling produces estimation error. Erlang-B assumes a loss system with Poisson call arrivals, independent call holding times and an infinite-source approximation. Rounding N and channels to integers also changes the result.
The likely scalable choice is a hybrid: N=7 fixed reuse as a robust baseline, sectoring to reduce interference, dynamic borrowing for temporary local imbalance, and selective splitting for sustained high demand. A pure N=3 layout has more channels per cell but fails the conservative SIR target in the simplified model; pure fixed assignment can waste channels when traffic is uneven.

System-Level Explanation

The theoretical equations provide reference values under clearly defined assumptions. The simulator should first reproduce these reference values before additional effects are introduced. For the N = 7 and n = 4 case, the six-equidistant-interferer model gives approximately 18.66 dB. If a simulated reference position uses the same six distances and equal transmit powers, its result should be close to this value.
The Erlang-B comparison should use exactly the same number of channels and offered traffic in both calculations. If the analytical routine uses 48 channels and 38 Erlangs, the simulated or

independently calculated blocking result should be close to 1.8 percent. Differences caused only by rounding should be small. A large difference suggests an input or implementation error.

Spatial simulation can produce a different average SIR because mobile locations are not all at the same distance from the serving site. Some locations may be close to the centre, while others may be near a cell edge or corner. The interferers also have different distances, so the total interference is not always equal to six times one common received-power value.

Shadowing and antenna patterns create further differences. Shadowing changes the received power on individual radio links, while sector antennas apply different gains according to direction. A simulator that includes these effects is expected to produce a distribution of values rather than the single reference value from the simplified equation.

Finite-tier simulation can omit distant interferers that would be included in a more complete model. The effect depends on how many tiers are simulated and on the path-loss exponent. The simulation should include enough neighbouring cells for the result to stabilise, and the report should state the number of tiers used.

Erlang-B itself is an approximation to a loss system. It assumes a particular arrival and call-holding model and does not directly represent every feature of modern packet data traffic. Handoff calls, priority rules, retransmissions and correlated event traffic may require additional modelling. These limitations should be stated when interpreting differences between theory and actual network measurements.

The strategy comparison should use a consistent set of assumptions and a common traffic distribution. If one strategy is evaluated with a different number of users or a different random seed, the comparison may be misleading. Use the same sample positions and demand profile wherever possible, then report both the analytical baseline and the simulation results.

The most useful conclusion is not that simulation must always equal theory. Rather, the simple model should agree with theory under the same ideal assumptions, and differences should become explainable when realistic features are added. This process validates the code and shows which real-world effects have the greatest influence on the design.

Further design considerations

The comparison should begin with a controlled reference case in which the simulator is configured to match the analytical assumptions exactly. This is a unit test for the model. Once the reference case agrees, features such as random mobile positions, shadowing, directional antennas and additional interference tiers can be enabled one at a time.

A difference between theory and simulation is not automatically an error. It may be caused by a more realistic model, but the reason must be demonstrated. For example, if the simulated 5th-percentile SIR is lower than the analytical reference, the report should examine mobile locations near the cell edge and the distances to the dominant interferers.

Simulation uncertainty should be reported when random samples are used. Repeating the experiment with different seeds can show whether the mean and percentile values are stable. If the result changes substantially between runs, more samples may be required or the traffic distribution may need to be represented more accurately.

The theory–simulation comparison should be performed separately for each strategy. Fixed reuse can be compared against the ideal cluster model; sectorisation should include antenna

gain; dynamic assignment should include the channel-selection rule; and cell splitting should include the changed cell geometry. A single SIR value cannot validate every part of the network model.

The final discussion should identify which assumptions have the greatest effect on the result. Possible factors include the equal-distance interferer approximation, the number of simulated tiers, shadowing, the distribution of mobile positions, antenna patterns, the Erlang-B traffic assumptions and the rounding of channel allocation. This makes the limitations explicit and supports a defensible recommendation.

Operational Analysis

The comparison should explain the purpose of each model. Analytical equations provide a simple and transparent prediction, while simulation allows more detailed geometry and traffic assumptions to be included. Neither method is automatically correct in every situation; both depend on the assumptions and implementation used.

When the simulation is extended, each new feature should be tested separately. First add mobile-position variation, then add shadowing, then add antenna patterns and additional interferer tiers. This sequence makes it easier to identify which feature causes a difference from the analytical reference.

The simulator should use consistent units and power calculations. Received powers should be summed in linear units before converting the final SIR to decibels. Mixing linear and decibel quantities is a common source of error and should be checked during code review.

The final comparison should state whether the simulation supports the recommendation, what limitations remain and what further measurements are needed. This is more meaningful than simply stating that the theoretical and simulated results are different.

Implementation and planning notes

The first comparison should use the same geometry and assumptions in both methods. If the theoretical equation assumes six equal-distance interferers, the reference simulation should use the same six interferer distances. This controlled comparison checks whether the formula and the code are consistent.

Once the reference case is verified, the model can be made more realistic. Mobile positions can be varied, antenna patterns can be included and shadowing can be added. Each extension should be documented so that the reader understands why the results move away from the ideal analytical value.

The simulator should use a sufficiently large number of samples to estimate the SIR distribution. The number of samples should be stated and, where possible, repeated runs should be used to check stability. This is particularly important for lower-percentile values because they are affected by the tail of the distribution.

The Erlang-B comparison should be checked independently of the SIR comparison. The two models describe different aspects of network performance. A program may calculate SIR correctly but have an error in the traffic allocation or blocking calculation, so both require separate validation.

The report should identify whether a difference is caused by a modelling assumption, a random sample or a programming error. This classification makes the discussion more useful and demonstrates that the simulator has been tested rather than simply executed.
The final strategy decision should be based on the combined evidence. The selected alternative should meet the radio-quality and traffic targets under the tested conditions, and its site and operating requirements should remain practical for the township.
The comparison is complete only when the reasons for agreement and disagreement are discussed. A close match under ideal assumptions verifies the implementation, while differences after adding realistic effects show how the assumptions influence performance. This is the main value of comparing theory with simulation.
The final conclusion should distinguish code validation from network validation. Matching a known equation confirms that a calculation has been implemented correctly, but it does not prove that the assumptions represent the real township. Field measurements are still required before deployment.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommended township design: use N=7 as the baseline reuse cluster, target SIR ≥18 dB, use three-sector (120°) antennas at high-interference/high-demand sites, and apply selective micro-cell splitting in dense hostels, academic blocks and road corridors. Use dynamic channel borrowing only when a borrowed channel passes the interference check. Begin with R≈0.5 km for the initial layout, then densify as traffic maps show overload.
For R=0.5 km, 19 cells approximately cover 12 km². Each N=7 cell has 48 channels and a planning limit near 38 Erlangs at 2% blocking, giving an idealised total of about 722 Erlangs. This is enough for the 1,000-user load (25 Erlangs) and, in a uniform ideal model, 10,000 users (250 Erlangs), but not 20,000 users (500 Erlangs) if coverage, busy-hour imbalance and reserve margins are considered; 100,000 users require roughly ceil(2500/38)=66 traffic-capable cells as an ideal lower bound. At R=0.25 km, about 74 cells cover 12 km² and the idealised capacity is about 2,812 Erlangs.
The design balances interference, capacity, handoffs and cost. Increasing N improves SIR but lowers channels per cell; sectoring improves spatial isolation but adds antennas; splitting adds sites and backhaul and increases handoffs. Beyond 100,000 users, a macro-only design is not practical without dense small cells, careful backhaul planning and continued traffic measurements. Final values must be validated by a propagation survey and detailed simulation.

Technical Background and Application

The recommended plan begins with N = 7 fixed reuse because it provides a useful balance between channels per cell and co-channel separation under the selected assumptions. With 336 traffic channels, this gives 48 channels per cell. For n = 4, the six-interferer approximation gives an SIR close to 18.66 dB, which is slightly above the selected 18 dB reference target.
The initial geometric radius of 0.5 km gives a hexagonal area of approximately 0.6495 km². Dividing the township area of 12 km² by this area gives an estimate of 19 cells. This is a

planning starting point and should be adjusted after site surveys, indoor coverage checks and boundary optimisation.

The approximate 38-Erlang planning load per 48-channel cell gives an idealised aggregate capacity of about 722 Erlangs for 19 cells. The total traffic estimates are 25 Erlangs for 1,000 subscribers, 250 Erlangs for 10,000, 500 Erlangs for 20,000 and 2,500 Erlangs for 100,000 subscribers. These aggregate comparisons assume uniform traffic and should be followed by per-cell loading checks.

Three-sector antennas should be introduced at sites where interference or concentrated demand justifies the extra equipment. The sector azimuths should be selected using the actual building layout and the location of neighbouring co-channel cells. Antenna tilt should be adjusted to control coverage overlap and avoid sending unnecessary energy into distant cells.

Dynamic channel borrowing should be used as a local optimisation, not as a replacement for a sufficient overall capacity plan. A borrowed channel must be free, compatible with the interference constraints and available without making the donor cell vulnerable to its own busy-hour demand. Borrowing is most useful when the traffic imbalance is temporary or varies by time of day.

Selective micro-cell splitting should be used in persistent hot spots. Candidate areas include hostels, lecture buildings, libraries, cafeterias and busy road corridors. Before installing a micro-cell, the operator should compare the cost of a new site with the expected reduction in blocking and the additional carried traffic. A micro-cell also requires power, backhaul, access permission and an updated neighbour plan.

The hybrid plan is superior to relying on only one method because each technique solves a different problem. Fixed reuse establishes predictable spectrum sharing; sectorisation reduces directional interference; borrowing improves resource utilisation; and cell splitting adds local capacity. The final choice should be supported by both analytical calculations and simulation outputs, with any unmeasured performance value clearly identified.

Beyond 100,000 subscribers, the macro-only layout is unlikely to remain sufficient under the stated traffic assumption. The operator should add small cells in high-demand zones, improve backhaul, monitor handoff performance and repeat the capacity analysis as the subscriber base grows. The network should be expanded in stages rather than installing the maximum number of sites before demand is demonstrated.

Further design considerations

The deployment should begin with a baseline survey and a coverage plan for the full 12 km² area. The initial radius and site count should be treated as planning values, then adjusted to match actual buildings, roads and terrain. Each site should have a documented location, antenna height, azimuth, tilt, backhaul route and expected coverage area.

After the baseline network is operating, performance should be reviewed using busy-hour channel occupancy, blocking, radio quality and handoff success. Cells that show high blocking but acceptable SIR may need additional capacity. Cells with poor SIR may need antenna or reuse optimisation. Cells with both problems may need a more substantial change such as a micro-cell.

The hybrid design should be implemented in stages so that the operator can measure the effect of each improvement. Sectorisation can be introduced at selected sites first, followed by dynamic borrowing and selective splitting where required. The results should be compared with the original baseline to confirm that the additional cost has produced the expected benefit.

Capacity planning should include reserve for future growth, but reserve should be based on the uncertainty in the forecast and the time required to install new sites. If site acquisition takes a long time, a larger planning margin may be appropriate. If the network can be expanded quickly, staged upgrades may reduce the amount of unused capacity.

The final design should be reviewed when the subscriber count approaches each planning milestone. The traffic estimate should be recalculated using the current subscriber number and the 0.025 Erlang assumption, then compared with actual busy-hour measurements. If observed traffic differs, the model should be updated before new capital expenditure is approved.

Design Evaluation

The final recommendation should include the selected cluster size, initial cell radius, channel allocation, SIR target, sectorisation policy and the conditions under which borrowing or splitting is used. Listing these parameters makes the proposed design specific enough to be evaluated and implemented.

The deployment plan should identify the locations where additional capacity is most likely to be required. The scenario names academic blocks, hostels, residential colonies and a main arterial road. The final site locations should be selected from a traffic map and propagation survey rather than assumed from the township average alone.

The operator should define a review process after each subscriber milestone. The review should compare forecast traffic with measured traffic and check blocking, SIR, handoff success and site availability. This keeps the design responsive to actual network behaviour.

The final decision should balance technical performance and cost. The design with the largest theoretical capacity may require unnecessary sites and operating expenditure. A hybrid strategy is preferred because it applies each method where its benefit is most relevant and preserves the fixed spectrum constraint.

Implementation and planning notes

The recommended design should be treated as a phased deployment. The first phase establishes the macro-cell coverage and the N = 7 frequency-reuse baseline. The next phase introduces sectorisation at sites where measurements show interference or concentrated demand. Later phases add borrowing and selective micro-cells where the traffic map demonstrates a need.

The operator should keep a record of the channel plan and antenna configuration at every site. This record supports fault diagnosis and future optimisation. When a new micro-cell is installed, its neighbour relationships and channel assignment should be added to the same planning record.

The design should include a process for checking the 2 percent blocking target. Cells approaching the threshold should be monitored more frequently, especially during the busy hour. If blocking exceeds the target, the operator should determine whether the cause is traffic growth, channel allocation, interference or equipment availability.

The plan should also include a process for checking the 18 dB SIR target. A cell may have adequate traffic channels but poor radio quality at the edge. Drive testing and indoor measurements can identify where antenna adjustment, sectorisation or a new site is required.
The recommendation should state the assumptions that could change the result. These include the subscriber traffic rate, path-loss exponent, cell radius, site cost, channel pool and event attendance. If any of these values changes significantly, the design should be reviewed rather than relying on the original calculation.
A hybrid approach is appropriate because it can grow with demand without changing the fixed spectrum allocation. It combines predictable reuse with local interference management and selective capacity addition, which is more flexible than relying on a single strategy for every part of the township.
The recommended hybrid plan should be documented in a form that can be updated. The channel groups, site locations, sector directions, cell radii and performance measurements should be retained for future planning. This information will help the operator respond to subscriber growth without repeating the entire design process.
The final recommendation is based on the fixed spectrum constraint and the need to balance interference, traffic capacity and cost. A staged hybrid network provides a practical way to maintain service as demand grows, while allowing local changes where the measurements show that they are necessary.

TASK 12 – Re-evaluate and Adapt the Design: Day-4 Challenge (8 Marks)

A. Impact Analysis (2 Marks)

Event traffic is 25,000 attendees × 0.05 Erlang/person = 1,250 Erlangs concentrated in 1 km² for four hours. The event traffic density is 1,250 Erlangs/km². For comparison, the 100,000-user township average is 2,500/12 = 208.3 Erlangs/km², so the stadium event density is about six times that average. Compared with the 20,000-user township average (500/12 = 41.7 Erlangs/km²), it is about 30 times higher.
The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. The event is localised, short, large and predictable, so a temporary solution can be more economical than permanently densifying the entire township.

B. Evaluation of the Original Design (2 Marks)

At R=0.25 km, cell area is 0.1624 km², so approximately 1/0.1624 = 6.16 cell-equivalents lie in the 1 km² stadium zone. At about 38 Erlangs per 48-channel cell, this gives only about 234 Erlangs of idealised traffic capacity, far below 1,250 Erlangs. The original design therefore cannot carry the event traffic with a 2% blocking target. Blocking and congestion would rise first; interference and access-channel congestion may also increase as users concentrate.
A simple lower-bound cell count is ceil(1250/38)=33 cells with 48 channels per cell and 2% blocking target. This excludes reserve capacity, traffic imbalance, control-channel limits and interference constraints, so an operational design should provide margin beyond 33 cell-equivalents.

C. Adaptation or Redesign (2 Marks)

Decision: MODIFY temporarily rather than permanently redesign the whole township. Deploy temporary cells-on-wheels or compact temporary micro-cells around the stadium, use sectorised antennas and interference-aware dynamic channel borrowing, and tune transmit power to avoid excessive overlap. Maintain N=7 as the starting reuse plan; adjust local channel groups and sector orientation after an event-zone interference check.

Plan for at least 33 idealised 48-channel cell-equivalents to meet 1,250 Erlangs at approximately 2% blocking; add a practical reserve based on measured load and outages. The temporary sites increase local capacity and improve SIR through shorter serving distances and sectoring, but they also increase handoffs and require backhaul, power, permits and on-site coordination. A planning budget should include event surveys, temporary equipment rental, installation, transport and four-hour operation. A permanent build-out would incur recurring cost despite events occurring only about twice a month.

D. Evidence-Based Defence (2 Marks)

Temporary modification is the best trade-off because the surge is predictable, concentrated in one square kilometre and lasts only four hours, about twice monthly. A permanent network sized for 1,250 Erlangs/km² would overbuild most of the township during normal days. Temporary cells-on-wheels and sectorised micro-cells put capacity where and when it is needed, while dynamic borrowing can use spare channels from neighbouring cells if interference permits.

The decision is supported by the calculation: 1,250 Erlangs are offered in the stadium zone, while approximately six existing cell-equivalents provide only about 234 Erlangs at the 2% target. At least 33 idealised cell-equivalents are needed before adding reserve margin. The revised design should be accepted only after the simulator verifies SIR/outage and blocking for the event traffic map. The main limitations are the simplified uniform traffic assumption, approximate Erlang-B model, uncertain shadowing, and the need to verify real antenna patterns and temporary-site backhaul.

Detailed Design Explanation

The event demand is calculated directly from the number of attendees and their offered traffic. With 25,000 attendees and 0.05 Erlang per attendee, the total offered traffic is 1,250 Erlangs. The traffic is concentrated into one square kilometre, so the relevant measure is not only total township traffic but also the traffic density in the stadium zone.

The event density is 1,250 Erlangs per square kilometre. At the 100,000-subscriber township stage, the average density is 2,500/12, or about 208.3 Erlangs per square kilometre. The stadium event density is therefore approximately six times the township-wide average at that stage. It is much higher than the average at the lower subscriber stages.

The event changes the original uniform traffic assumption. The attendees arrive in a limited area and generate demand over a short period. The network must also handle access attempts and signalling activity that may occur in bursts, such as when attendees enter the venue or leave

after the event. A design based only on the average daily load will not describe this peak condition.

Using R = 0.25 km, the ideal hexagonal cell area is approximately 0.1624 km². Dividing one square kilometre by this area gives about 6.16 cell-equivalents in the stadium zone. At 38 Erlangs per 48-channel cell, the idealised capacity is approximately 234 Erlangs. This is far below the event demand of 1,250 Erlangs.

The lower-bound capacity calculation gives 1,250/38, which is approximately 32.9, so at least 33 idealised cell-equivalents are required before any reserve margin is added. This is not a final site count because it assumes uniform loading, identical cell capacity and no additional interference or signalling limitation. The operational plan should provide a margin based on event measurements and simulation.

Temporary cells-on-wheels or compact temporary micro-cells are suitable because the event is predictable and occurs only about twice each month. The operator can deploy additional capacity before the event and remove or redeploy it afterwards. This avoids permanently building the entire township network for a short-lived peak that is localised to the stadium.

Sectorised antennas can direct radio resources into the audience area and reduce unnecessary overlap with surrounding cells. The antenna azimuth and tilt should be selected using the venue layout, stand positions, entrances, exits and temporary site locations. The plan should also check that the temporary cells do not create strong interference for existing macro-cells or neighbouring sectors.

Dynamic channel borrowing can supplement the temporary capacity if nearby cells have spare eligible channels. It should not be treated as the main solution because the stadium demand is much larger than the estimated capacity of the existing cell-equivalents. Borrowing can only redistribute idle resources; it cannot create additional spectrum or solve a network-wide shortage.

The temporary event plan should include radio surveys, temporary site rental, transport, installation, power, backhaul, permits, monitoring and removal. The operator should test the equipment before the event and monitor blocking, access failures, SIR and handoff performance during the event. A contingency plan should be available if attendance or traffic exceeds the forecast.

The final decision is to modify the network temporarily rather than permanently redesign the whole township. This decision follows from the four-hour duration, the expected frequency of events and the large difference between event demand and the capacity of the original stadium-zone cell-equivalents. The plan should be confirmed through simulation and site measurements before event-day deployment.

Further design considerations

The event analysis should separate the four-hour peak from the normal days of operation. The temporary network should be planned for the event period, while the permanent macro network should continue to serve the surrounding township. This separation makes it possible to add capacity where it is needed without changing the entire frequency plan permanently.

The stadium coverage map should identify the audience seating area, open ground, entrances, exits, service areas and nearby buildings. User density may not be uniform across the one-

square-kilometre zone. Temporary cells should be placed according to the expected concentration of users, and the coverage boundaries should be checked to avoid creating weak areas or excessive overlap.

Event traffic may arrive in bursts. For example, many devices may attempt to register or send messages shortly after attendees enter the stadium. The network should therefore check control-channel and access capacity as well as traffic-channel capacity. The 1,250-Erlang calculation describes offered traffic but does not by itself establish that signalling resources are sufficient.

Temporary sites require reliable backhaul. If a temporary cell has radio capacity but cannot connect that traffic to the core network, the additional capacity cannot be used effectively. Backhaul should be sized for the expected event load and tested before the event. Power backup and physical security should also be included in the event plan.

After the event, the operator should review the measured traffic, blocking, SIR, access failures and handoff activity. These measurements can be compared with the forecast and used to improve the plan for the next event. Since events occur about twice each month, a repeatable deployment and monitoring procedure can reduce preparation time and operational risk.

Engineering Discussion

The temporary plan should be prepared before the event date and should include a deployment schedule. Temporary equipment must be transported, installed, configured, tested and connected to backhaul before attendees arrive. The removal plan should also be included so that temporary infrastructure does not remain unnecessarily after the event.

The event-day monitoring team should watch channel occupancy, blocking, radio quality, access failures and handoff performance. If one temporary cell becomes overloaded, traffic may need to be balanced across sectors or nearby cells where safe. Any change should be made using the interference plan rather than by assigning channels without checking compatibility.

The operator should compare actual event traffic with the forecast of 1,250 Erlangs. If the measured value is lower, the temporary capacity plan may be reduced for future events. If it is higher, the number or placement of temporary sites may need to be revised. This feedback makes later event deployments more accurate.

The final defence should connect the duration and frequency of the event to the investment decision. Since the peak lasts four hours and occurs roughly twice a month, temporary capacity is likely to be more economical than permanently sizing every township cell for the stadium peak. The decision remains subject to the site survey, interference checks and cost quotations.

Implementation and planning notes

The event plan should be coordinated with the stadium organiser because the expected audience distribution depends on seating, entrances, exits and event schedules. The operator should obtain an event layout and use it to identify the areas where temporary radio capacity is required. A uniform placement of temporary sites may not match the audience density.

Temporary cells should be tested under load before the event. The test should verify radio coverage, channel assignment, backhaul throughput, power backup and handoff between temporary and permanent cells. Any interference problem should be corrected before the audience arrives.

The event plan should include a contact and escalation procedure for the operations team. If a temporary site fails, traffic may need to be handled by neighbouring cells or another temporary unit. The team should know which configuration changes are permitted and how to monitor their effect.

The 1,250-Erlang estimate is the offered traffic under the assignment's per-attendee assumption. Actual event traffic may vary according to the type of event, audience behaviour and time of day. The operator should compare the forecast with measured event data after each convocation or sports event.

Because events occur roughly twice per month, temporary equipment can be scheduled and reused. The operator may keep a repeatable kit and deployment procedure rather than purchasing permanent capacity for the stadium zone. The cost comparison should include rental or ownership, transport, installation, backhaul and maintenance.

The final choice remains a temporary modification of the original network. This is supported by the high local traffic density and short event duration. Permanent township-wide redesign would add capacity where it is not needed during normal operation, while temporary cells and sectorisation target the event area and time.

The event plan should be reviewed after every event because attendance, device activity and audience distribution may change. The operator should retain the event measurements and compare them with the original forecast. This evidence can improve the temporary site count and placement for future events.

The temporary approach avoids permanently increasing the capacity of the entire township for a short, localised surge. It focuses investment on the stadium zone during the four-hour event period, while the permanent network continues to serve normal township demand. This is the basis for the recommended modification.

Overall conclusion

A fixed spectrum can support a growing township through spatial reuse, sectorisation, dynamic channel assignment and selective cell splitting. N=7 provides an approximately 18.7 dB reference SIR when n=4 under the six-interferer approximation, while 48 channels per cell support about 38 Erlangs at 2% blocking. Traffic growth requires additional cells and careful reuse planning; the stadium surge requires temporary local capacity rather than permanent township-wide overbuilding. All final site counts and cost figures should be refined using field surveys, measured traffic and a full spatial simulation.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Design Alternative Frequency Planning Strategies	5
Task 2 – Identify the Governing Design Parameter	2
Task 3 – Derive the Capacity and Interference Functions Mathematically	6
Task 4 – Analyse Best, Average and Worst Case Behaviour	4
Task 5 – Scaling and Sensitivity Analysis	3
Task 6 – Formulate the Recurrence for Repeated Cell Splitting	3
Task 7 – Solve the Recurrence and Interpret the Result	4
Task 8 – Implement and Test the Planning Strategies	4
Task 9 – Record and Present Experimental Performance	3
Task 10 – Compare Theory with Simulation	4
Task 11 – Recommend and Defend the Final Design	4
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Task 12 – Day-4 Design Challenge	8
TOTAL	**50**